

With ordinary gradient descent, the following are the relevant objective and gradient functions for LS regression:

$$p(\alpha, \beta; P) = \sum_{u \in P} (y_u - \alpha - \beta x_u)^2$$

$$g(\alpha, \beta; P) = -2 \begin{bmatrix} \sum_{u \in P} (y_u - \alpha - \beta x_u) \\ \sum_{u \in P} x_u (y_u - \alpha - \beta x_u) \end{bmatrix}$$

These functions are calculated over all of the data in the population  $P$ .

With batch-sequential gradient descent, these functions are identical with respect to  $\alpha$  and  $\beta$ . The only difference is that we use a subset of the data to calculate them. Call this subset batch  $k$ , and denote it  $B_k \subset P$ . The relevant objective and gradient functions are therefore:

$$p(\alpha, \beta; B_k) = \sum_{u \in B_k} (y_u - \alpha - \beta x_u)^2$$

$$g(\alpha, \beta; B_k) = -2 \begin{bmatrix} \sum_{u \in B_k} (y_u - \alpha - \beta x_u) \\ \sum_{u \in B_k} x_u (y_u - \alpha - \beta x_u) \end{bmatrix}$$

These functions are calculated over a subset  $B_k$  of the data in population  $P$ .

these are approximations of  $p(\alpha, \beta; P)$  and  $g(\alpha, \beta; P)$ , respectively

With batch-stochastic gradient descent, these functions are again identical with respect to  $\alpha$  and  $\beta$ , and differ only with respect to which data is plugged into them. In this case we calculate these functions on a sample  $S \subset P$  randomly drawn from the population  $P$ . The relevant objective and gradient functions are therefore:

$$p(\alpha, \beta; S) = \sum_{u \in S} (y_u - \alpha - \beta x_u)^2$$

$$g(\alpha, \beta; S) = -2 \begin{bmatrix} \sum_{u \in S} (y_u - \alpha - \beta x_u) \\ \sum_{u \in S} x_u (y_u - \alpha - \beta x_u) \end{bmatrix}$$

These functions are calculated over a sample  $S$  from the data in population  $P$ .

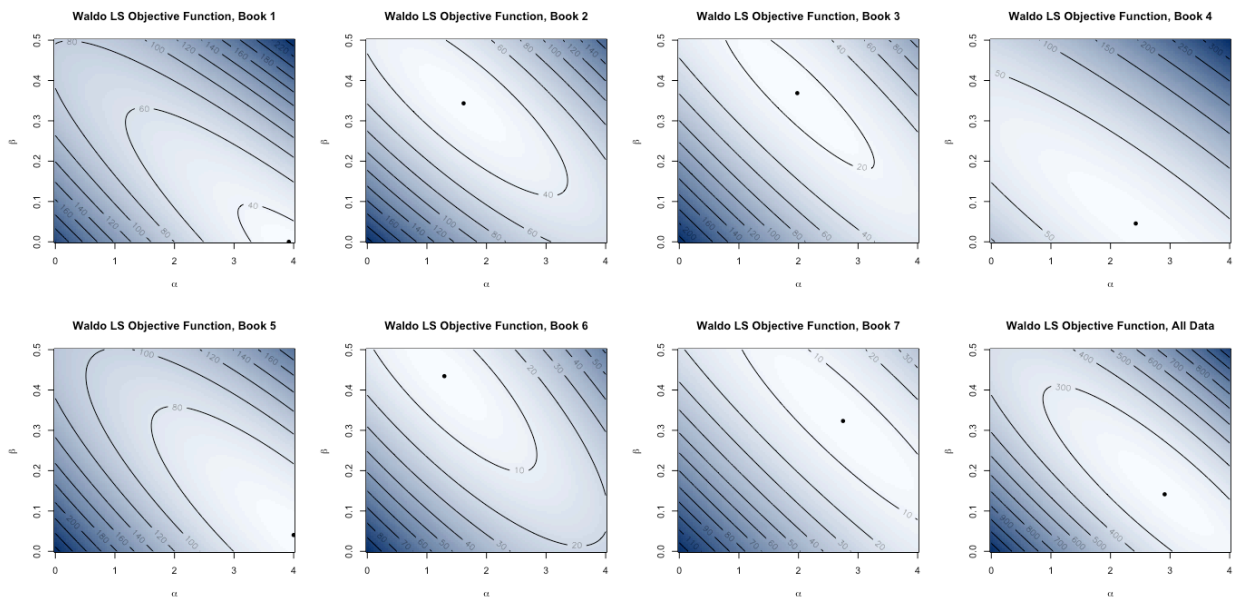
these are approximations of  $p(\alpha, \beta; P)$  and  $g(\alpha, \beta; P)$ , respectively

To illustrate the impact calculating these functions over different subsets of the data has, let's consider the waldo data. Below we visualize (via contour plots) the objective function with different data being plugged in.

## Situation 1: Batch - Sequential GD

Suppose that we partition the population into 7 batches corresponding to the 7 Waldo books:

$B_1, B_2, \dots, B_K$ . In batch-sequential gradient descent will use a different batch's (book's) data at each step in the algorithm. As noted above, what we're doing is approximating  $p(\alpha, \beta; P)$  and hence  $g(\alpha, \beta; P)$  at each step. Below I visualized these approximations. The bottom-right contour visualizes the true objective function  $p(\alpha, \beta; P)$  based on all of the data. The other 7 contours show the approximations of this based on batches (books) 1-7.



The variation in these surfaces illustrates the source of noise in our solution path. Suppose we start at  $(\alpha, \beta) = (0, 0)$  and start walking toward the minimum, when we use  $p(\alpha, \beta; P)$  to determine our gradient and hence directions, we'll get to the minimum efficiently. But when we use  $p(\alpha, \beta; B_k)$  for  $k=1, 2, \dots, 7$  we'll occasionally move in suboptimal directions.

### Situation 2: Batch-Stochastic GD

An analogous illustration is shown below for batch-stochastic gradient descent. Again, the bottom-right contour visualizes the true objective function  $p(\alpha, \beta; P)$  based on all of the data. The other 7 contours show the approximations based on seven different samples (of size  $n=12$ ) of the waldo data. We draw similar conclusions as above, except that these approximations appear better than those in batch-sequential and therefore the corresponding gradients will behave more like the true gradient  $g(\alpha, \beta; P)$ .

